

### Knowledge Engineerin

#### Algorithms for Artificial Intelligence and Graphs

##### IDENTIFICATION

CODE : IF-3-AAIA  
ECTS : 3.0

##### HOURS

Lectures : 13.5 h  
Seminars : 12.0 h  
Laboratory : 12.0 h  
Project : 0.0 h  
Teacher-student  
contact : 37.5 h  
Personal work : 35.0 h  
Total : 72.5 h

##### ASSESSMENT METHOD

DS and TP

##### TEACHING AIDS

<http://liris.cnrs.fr/peportie/#enseignements>  
<http://liris.cnrs.fr/csolnon/supportAlgoGraphes.pdf>  
Platform for running algorithms  
(based on DomJudge)

##### TEACHING LANGUAGE

French

##### CONTACT

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##### AIMS

A first goal is to deepen some principles and practice introduced during the first semester, in the course IF-3-ALGO : Specification of an algorithm by means of pre- and post-conditions, Algorithm complexity, Proof of correction of an algorithm, etc.

A second goal is to study some classical algorithm coming from two (non-disjoint) domains: Graph theory and Artificial Intelligence

Targeted skills:

- Choose relevant algorithms and data structures, and evaluate their complexity
- Master recursion and iteration
- Recognize the complexity class of a problem, and choose an appropriate solution process with respect to this class
- Analyze a decision making problem

##### CONTENT

- Graph Algorithms: Definitions, Data structures for modelling graphs, Graph traversals, Shortest path problems
- Planning Algorithms: Modelling a problem as the search of a plan in a state-transition graph, Looking for optimal paths in a state-transition graph
- Introduction to computational complexity: Complexity classes, Problem reduction
- Algorithms for solving NP-hard problems: Notion of search space, Exhaustive exploration of a search space (Branch and Bound, Minimax/alpha-beta, A\*), heuristic and meta-heuristic approaches (greedy construction, local search and simulated annealing, neural networks)

Practice through exercises, with an emphasis on the traveling salesman problem

##### BIBLIOGRAPHY

- Introduction à l'algorithmique - Cormen, Leiserson, Rivest  
Edelkamp, Stefan, and Stefan Schroedl. Heuristic search: theory and applications. Elsevier, 2011.
- RUSSELL, NORVIG, artificial intelligence a modern approach, 3rd edition, 2010
- Haykin, Simon S., et al. Neural networks and learning machines. Vol. 3. Upper Saddle River: Pearson Education, 2009.
- Michalewicz, Zbigniew, and David B. Fogel. How to solve it: modern heuristics. Springer Science & Business Media, 2004.
- Negnevitsky, Michael. Artificial intelligence: a guide to intelligent systems. Pearson Education, 2005.
- BACKHOUSE, program construction the correct way, 2002
- Langville, Amy N., and Carl D. Meyer. Google's PageRank and beyond: The science of search engine rankings. Princeton University Press, 2011.

##### PRE-REQUISITE

IF-3-ALGO

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